

1. What is Data Science?

Data Science is an interdisciplinary field that combines:

- Statistics
- Mathematics
- Programming
- Machine Learning
- Artificial Intelligence
- Data Analysis

to extract meaningful insights and knowledge from structured and unstructured data.

The main goal of Data Science is to solve real-world problems using data-driven techniques.

Example:

Netflix recommends movies based on user behavior using Data Science.

Main Components:

- Data Collection
 - Data Cleaning
 - Data Analysis
 - Visualization
 - Machine Learning
 - Decision Making
-

2. Why is Data Science needed?

Data Science is needed because huge amounts of data are generated daily from social media, banking, healthcare, e-commerce, etc.

It helps organizations:

- Make better decisions
- Predict future trends
- Increase profits
- Improve customer experience
- Automate tasks
- Detect fraud and risks

Example:

Amazon uses Data Science to recommend products to customers.

3. List applications of Data Science.

Applications include:

Healthcare

- Disease prediction
- Medical image analysis

Finance

- Fraud detection
- Stock market prediction

E-commerce

- Product recommendations
- Customer segmentation

Social Media

- Sentiment analysis
- Personalized feeds

Transportation

- Self-driving cars
- Traffic prediction

Cybersecurity

- Threat detection
- Spam filtering

Weather Forecasting

- Climate prediction

Education

- Student performance analysis
-

4. What is the relationship between Data Science and Information Science?

Information Science focuses on:

- Storing information
- Managing information

- Retrieving information

Data Science focuses on:

- Analyzing data
- Finding patterns
- Predicting outcomes

Relationship:

Information Science provides methods to organize and manage data, while Data Science uses that data for analysis and intelligent decision-making.

5. Differentiate between Business Intelligence and Data Science.

Business Intelligence	Data Science
Focuses on past data	Focuses on future predictions
Generates reports and dashboards	Uses machine learning and AI
Descriptive analysis	Predictive and prescriptive analysis
Mostly structured data	Structured + unstructured data
Answers “What happened?”	Answers “What will happen?”

Example:

- BI: Monthly sales report
- Data Science: Predict future sales

6. What are different types of data?

1. Structured Data

Data stored in rows and columns.

Example:

Excel sheets, SQL databases

2. Unstructured Data

Data without a fixed format.

Example:

Images, videos, audio, social media posts

3. Semi-Structured Data

Data with partial organization.

Example:

JSON, XML

4. Qualitative Data

Descriptive data.

Example:

Color, gender, feedback

5. Quantitative Data

Numerical data.

Example:

Age, salary, marks

7. What is data collection?

Data collection is the process of gathering information from different sources for analysis.

Sources:

- Surveys
- Sensors
- Databases
- APIs
- Websites
- IoT devices

Types:

- Primary Data: Collected directly
- Secondary Data: Already available

Importance:

Accurate data collection improves analysis quality.

8. What is Data Wrangling?

Data Wrangling is the process of:

- Cleaning
- Organizing
- Transforming
- Structuring

raw data into a usable format.

It is also called **Data Munging**.

Tasks involved:

- Removing duplicates
- Handling missing values
- Formatting data
- Merging datasets

Example:

Converting date formats into a standard format.

9. Why is Data Wrangling needed?

Raw data is often:

- Incomplete
- Noisy
- Inconsistent
- Incorrect

Data Wrangling improves:

- Data quality
- Accuracy
- Reliability
- Machine learning performance

Without wrangling:

Wrong insights and poor predictions may occur.

10. What is Data Cleaning?

Data Cleaning is the process of detecting and correcting errors in datasets.

It includes:

- Removing duplicates
- Handling missing values
- Correcting formats
- Removing invalid records

Example:

Replacing empty age values with average age.

Importance:

Clean data gives accurate results.

11. What is Data Integration?

Data Integration combines data from multiple sources into one unified dataset.

Sources may include:

- Databases
- APIs
- Excel files
- Cloud storage

Example:

Combining customer data from sales and marketing departments.

Benefits:

- Better analysis
 - Centralized information
 - Improved decision-making
-

12. What is Data Reduction?

Data Reduction decreases the size of data while preserving important information.

Techniques:

- Sampling
- Aggregation
- Feature selection

- Dimensionality reduction

Benefits:

- Faster processing
 - Reduced storage
 - Better performance
-

13. What is Data Transformation?

Data Transformation converts data into a suitable format for analysis.

Examples:

- Scaling
- Encoding
- Normalization
- Aggregation

Example:

Converting “Male/Female” into 1/0.

Purpose:

Improves compatibility with machine learning algorithms.

14. What is Data Discretization?

Data Discretization converts continuous data into intervals or categories.

Example:

Age:

- 0–18 → Child
- 19–60 → Adult
- 60+ → Senior Citizen

Advantages:

- Simplifies analysis
 - Improves model understanding
-

15. Why is statistics important in Data Science?

Statistics helps in:

- Data analysis
- Prediction
- Decision-making
- Finding patterns
- Hypothesis testing

Uses:

- Understanding data distribution
- Measuring relationships
- Building ML models

Example:

Mean and standard deviation summarize data.

16. What is Mean?

Mean is the average value of data.

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Where:

- $\sum x$ = Sum of observations
- n = Number of observations

Example:

Data = 2, 4, 6

Mean = $(2+4+6)/3 = 4$

17. What is Median?

Median is the middle value in sorted data.

Steps:

1. Arrange data in order
2. Find middle value

Example:

2, 4, 6, 8, 10

Median = 6

Importance:

Less affected by outliers.

18. What is Mode?

Mode is the most frequently occurring value in a dataset.

Example:

1, 2, 2, 3, 4

Mode = 2

Use:

Useful for categorical data.

19. What is Mid-range?

Mid-range is the average of the highest and lowest values.

Formula

$$\text{Mid-range} = \frac{\text{Maximum} + \text{Minimum}}{2}$$

Example:

Minimum = 10

Maximum = 50

Mid-range = $(10 + 50)/2 = 30$

20. What is Range?

Range is the difference between maximum and minimum values.

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example:

Max = 50

Min = 10

Range = 40

Importance:

Measures data spread.

21. What is Variance?

Variance measures how far data points are spread from the mean.

Low Variance:

Data points are close to mean.

High Variance:

Data points are widely spread.

Formula

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Where:

- μ = Mean
- N = Number of observations

22. What is Standard Deviation?

Standard deviation is the square root of variance.

It measures dispersion in the dataset.

Formula

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Importance:

- Low SD → Data close to mean
- High SD → Data widely spread

23. What is Mean Deviation?

Mean deviation is the average of absolute deviations from the mean.

Formula

$$\text{Mean Deviation} = \frac{\sum |x - \bar{x}|}{n}$$

Purpose:

Measures variability in data.

24. What is Bayes' theorem?

Bayes' theorem calculates conditional probability.

Formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \approx 0.68, P(B) \approx 0.25$$

$P(B)=0.25$
 $P(B|A)P(A)=0.17$
 $P(A|B)\sim 0.68$
Posterior = useful evidence / total evidence

Where:

- $P(A|B)$ = Probability of A given B
- $P(B|A)$ = Probability of B given A

Applications:

- Spam filtering
 - Medical diagnosis
 - Machine learning
-

25. What is hypothesis?

A hypothesis is a testable assumption or statement about a population.

Example:

"Students who study more score higher marks."

26. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether a hypothesis is true.

Steps:

1. Define hypotheses
 2. Choose significance level
 3. Perform statistical test
 4. Accept or reject hypothesis
-

27. What is Null Hypothesis?

Null Hypothesis (H_0) states:

“There is no significant relationship or difference.”

Example:

“There is no difference in performance between two teaching methods.”

28. What is Alternative Hypothesis?

Alternative Hypothesis (H_1) states:

“There is a significant relationship or difference.”

Example:

“There is a difference in performance between two teaching methods.”

29. What is Pearson Correlation?

Pearson Correlation measures linear relationship between two variables.

Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Range:

- +1 → Perfect positive correlation
 - 0 → No correlation
 - -1 → Perfect negative correlation
-

30. What is a t-test?

A t-test checks whether means of two groups are significantly different.

Types:

- One-sample t-test
- Independent t-test
- Paired t-test

Example:

Comparing marks of two classes.

31. What is Chi-Square test?

Chi-Square test checks association between categorical variables.

Example:

Relationship between gender and product preference.

Uses:

- Feature selection
 - Hypothesis testing
-

32. What is Data Analytics Life Cycle?

The Data Analytics Life Cycle is a structured process used to analyze data and generate insights.

Phases:

1. Discovery
 2. Data Preparation
 3. Model Planning
 4. Model Building
 5. Communication
 6. Operationalize
-

33. What are the phases of Data Analytics Life Cycle?

1. Discovery

Understanding business problem.

2. Data Preparation

Cleaning and organizing data.

3. Model Planning

Choosing techniques and algorithms.

4. Model Building

Training models.

5. Communication

Presenting insights.

6. Operationalize

Deploying model in real environment.

34. What is the Discovery phase?

Discovery phase identifies:

- Business objectives
- Data sources
- Project requirements
- Risks

Output:

Clear understanding of the problem.

35. What is Data Preparation phase?

Data Preparation phase includes:

- Data cleaning
- Integration
- Transformation
- Formatting

Goal:

Prepare high-quality data for modeling.

36. What is Model Planning phase?

In this phase:

- Analytical methods are selected
- Algorithms are chosen
- Data relationships are studied

Example:

Choosing regression or classification.

37. What is Model Building phase?

In this phase:

- Models are trained
- Algorithms are implemented
- Performance is tested

Example:

Training a machine learning model using Python.

38. What is Communication phase?

Results are communicated using:

- Reports
- Graphs
- Dashboards
- Presentations

Goal:

Help stakeholders understand findings.

39. What is Operationalize phase?

Operationalize phase deploys the model into real-world systems.

Activities:

- Deployment
- Monitoring
- Maintenance
- Performance tracking

Example:

Deploying fraud detection system in banking.

40. Why is Data Analytics Life Cycle important?

It is important because it:

- Provides structured workflow
- Improves efficiency
- Reduces errors
- Ensures better decision-making
- Helps manage large projects effectively

Benefits:

- Better planning

- Better analysis
- Better deployment
- Reliable outcomes

40. Why is Data Analytics Life Cycle important?

The Data Analytics Life Cycle provides a systematic framework for handling data analysis projects efficiently.

Importance:

- Organizes the entire analytics process
- Improves accuracy and efficiency
- Reduces project risks
- Helps in better decision-making
- Ensures proper deployment of models

Benefits:

- Better project management
- Improved data quality
- Faster problem-solving
- Better communication among teams

Example:

Companies like Amazon and Netflix use analytics life cycles to improve recommendation systems.

41. What are essential Python libraries used in Data Science?

Python libraries simplify data analysis and machine learning tasks.

Important Libraries:

1. NumPy

Used for numerical computations and arrays.

2. Pandas

Used for data manipulation and analysis.

3. Matplotlib

Used for data visualization.

4. Seaborn

Used for statistical visualization.

5. Scikit-learn

Used for machine learning algorithms.

6. TensorFlow

Used for deep learning and AI.

7. Keras

High-level deep learning library.

8. SciPy

Used for scientific computations.

42. What is Data Preprocessing?

Data preprocessing is the process of preparing raw data for analysis and machine learning.

Steps involved:

- Data cleaning
- Handling missing values
- Encoding categorical variables
- Normalization
- Feature scaling

Importance:

- Improves model accuracy
- Removes noise and inconsistencies
- Makes data suitable for algorithms

Example:

Replacing missing salary values with average salary.

43. How do you remove duplicates in a dataset?

Duplicate records are removed using pandas.

Python Code:

```
df.drop_duplicates()
```

Why remove duplicates?

- Avoids incorrect analysis
- Prevents bias in machine learning models

- Improves data quality
-

44. How do you handle missing data?

Missing data can be handled using different techniques.

Methods:

1. Remove rows/columns

```
df.dropna()
```

2. Replace with mean

```
df['Age'].fillna(df['Age'].mean())
```

3. Replace with median or mode

4. Forward fill / Backward fill

Importance:

Proper handling improves model performance.

45. What is data transformation using mapping?

Mapping replaces existing values with new values using a dictionary.

Example:

```
df['Gender'] = df['Gender'].map({'Male':1,'Female':0})
```

Purpose:

- Converts categorical data into numerical format
 - Simplifies machine learning processing
-

46. What are types of analytics?

1. Descriptive Analytics

Analyzes past data.

Example:

Monthly sales report.

2. Diagnostic Analytics

Explains why something happened.

Example:

Why sales decreased.

3. Predictive Analytics

Predicts future outcomes.

Example:

Stock price prediction.

4. Prescriptive Analytics

Suggests actions for best outcomes.

Example:

Recommendation systems.

47. What is Predictive Analytics?

Predictive analytics uses historical data and machine learning to predict future events.

Techniques used:

- Regression
- Machine learning
- Data mining

Applications:

- Weather forecasting
- Fraud detection
- Demand forecasting

Example:

Predicting house prices.

48. What is Descriptive Analytics?

Descriptive analytics summarizes historical data to understand past events.

Techniques:

- Reports
- Dashboards
- Charts

- Aggregation

Example:

Sales summary of previous year.

Purpose:

Answers:

“What happened?”

49. What is Prescriptive Analytics?

Prescriptive analytics suggests optimal actions based on predictions.

Uses:

- Decision-making
- Optimization
- Recommendation systems

Example:

Google Maps suggesting fastest route.

Purpose:

Answers:

“What should be done?”

50. What is Apriori Algorithm?

Apriori is an association rule mining algorithm used to find frequent itemsets.

Applications:

- Market basket analysis
- Product recommendations

Example:

If customers buy bread and butter, they may also buy milk.

Key Concepts:

- Support
 - Confidence
 - Association Rules
-

51. What is FP-Growth Algorithm?

FP-Growth (Frequent Pattern Growth) is an efficient algorithm for finding frequent itemsets.

Advantages over Apriori:

- Faster execution
- No candidate generation
- Uses FP-Tree structure

Applications:

- Recommendation systems
 - Market analysis
-

52. What is Linear Regression?

Linear Regression is a supervised learning algorithm used to predict continuous values.

Formula

$$y = mx + b$$

m

b

-10-8-6-4-2246810-10-5510y-interceptx-intercept

Where:

- y = dependent variable
- x = independent variable
- m = slope
- b = intercept

Example:

Predicting house prices based on area.

Types:

- Simple Linear Regression
 - Multiple Linear Regression
-

53. What is Logistic Regression?

Logistic Regression is a classification algorithm used for binary outcomes.

Example:

- Spam or Not Spam

- Pass or Fail

Output:

Probability between 0 and 1.

Applications:

- Disease prediction
- Fraud detection

54. What is Naïve Bayes classifier?

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem.

Formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \approx 0.68, P(B) \approx 0.25$$

$P(B)=0.25$
 $P(B | A)P(A)=0.17$
 $P(A | B) \sim 0.68$
 Posterior = useful evidence / total evidence

Assumption:

Features are independent.

Applications:

- Spam filtering
- Sentiment analysis
- Text classification

55. What is Decision Tree?

Decision Tree is a supervised learning algorithm used for classification and regression.

Structure:

- Root Node
- Decision Nodes
- Leaf Nodes

Advantages:

- Easy to understand
- Handles categorical and numerical data

Applications:

- Loan approval
 - Medical diagnosis
-

56. What is Scikit-learn?

[Scikit-learn](#) is a Python machine learning library.

Features:

- Classification
- Regression
- Clustering
- Preprocessing
- Model evaluation

Advantages:

- Easy to use
 - Large community support
-

57. What is Clustering?

Clustering is an unsupervised learning technique that groups similar data points together.

Goal:

Maximize similarity within clusters and minimize similarity between clusters.

Applications:

- Customer segmentation
 - Image segmentation
 - Recommendation systems
-

58. What is K-Means clustering?

K-Means is an unsupervised clustering algorithm.

Working:

1. Select K clusters

2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroids
5. Repeat until convergence

Applications:

- Market segmentation
 - Pattern recognition
-

59. What is Hierarchical clustering?

Hierarchical clustering builds clusters in a tree-like structure.

Types:

1. Agglomerative

Bottom-up approach.

2. Divisive

Top-down approach.

Output:

Dendrogram

Applications:

- Biological classification
 - Document clustering
-

60. What is Time Series analysis?

Time Series Analysis studies data collected over time.

Components:

- Trend
- Seasonality
- Cyclic patterns
- Noise

Applications:

- Stock prediction
- Weather forecasting

- Sales forecasting
-

61. What is Text Analysis?

Text Analysis extracts useful information from text data.

Techniques:

- Sentiment analysis
- Topic modeling
- Keyword extraction

Applications:

- Chatbots
 - Social media analysis
 - Spam filtering
-

62. What is text preprocessing?

Text preprocessing cleans text before analysis.

Steps:

- Lowercasing
- Removing punctuation
- Stop-word removal
- Tokenization
- Stemming
- Lemmatization

Importance:

Improves NLP model performance.

63. What is Bag of Words?

Bag of Words converts text into numerical vectors based on word frequency.

Process:

1. Create vocabulary
2. Count word frequency

Example:

Sentence:

"I love AI"

Vocabulary:

[I, love, AI]

Vector:

[1,1,1]

64. What is TF-IDF?

TF-IDF stands for:

Term Frequency – Inverse Document Frequency.

Purpose:

Measures importance of words in a document.

Formula

$$TF-IDF = TF \times IDF$$

Applications:

- Search engines
 - Document ranking
 - NLP
-

65. What is Social Network Analysis?

Social Network Analysis studies relationships and interactions between people or organizations.

Components:

- Nodes
- Edges

Applications:

- Facebook friend analysis
 - Marketing
 - Fraud detection
-

66. What is Business Analytics?

Business Analytics uses data analysis and statistical methods to improve business decisions.

Types:

- Descriptive
- Predictive
- Prescriptive

Applications:

- Customer analysis
- Financial analysis
- Risk management

67. What is Model Evaluation?

Model evaluation measures performance of machine learning models.

Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Importance:

Helps choose best model.

68. What is Confusion Matrix?

Confusion Matrix compares actual and predicted values.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

69. What is Accuracy?

Accuracy measures overall correctness of predictions.

Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Example:

If 90 out of 100 predictions are correct:

Accuracy = 90%

70. What is Precision and Recall?

Precision

Measures correctness of positive predictions.

Formula

$$Precision = \frac{TP}{TP + FP}$$

Recall

Measures ability to identify all positive cases.

Formula

$$Recall = \frac{TP}{TP + FN}$$

Example:

Used in disease prediction systems.

71. What is AUC-ROC curve?

ROC Curve:

Plots True Positive Rate vs False Positive Rate.

AUC:

Area Under Curve.

Interpretation:

- AUC = 1 → Perfect model
- AUC = 0.5 → Random model

Applications:

Evaluating classification models.

72. What is Holdout method?

Holdout method splits dataset into:

- Training set
- Testing set

Common Split:

- 80% Training
- 20% Testing

Purpose:

Evaluate model performance.

73. What is Random Subsampling?

Random Subsampling repeatedly splits data into random train-test sets.

Advantage:

More reliable evaluation than single holdout.

Also called:

Repeated Holdout Method.

74. What is Parameter Tuning?

Parameter tuning optimizes model parameters for better performance.

Methods:

- Grid Search
- Random Search

Example:

Finding best K value in KNN.

75. What is Elbow Method?

Elbow Method determines optimal number of clusters in K-Means.

Process:

1. Calculate WCSS for different K values

2. Plot graph
3. Select elbow point

Purpose:

Choose best cluster count.

76. What is Data Visualization?

Data visualization represents data graphically.

Benefits:

- Easier understanding
- Pattern identification
- Better communication

Tools:

- Tableau
 - Power BI
 - Matplotlib
-

77. What are types of data visualization?

Types:

- Bar Chart
- Pie Chart
- Line Graph
- Scatter Plot
- Histogram
- Heatmap
- Box Plot

Each visualization is used for different purposes.

78. What are common visualization techniques?

1. Bar Chart

Compares categories.

2. Pie Chart

Shows percentage distribution.

3. Histogram

Displays frequency distribution.

4. Scatter Plot

Shows relationship between variables.

5. Line Plot

Displays trends over time.

79. What tools are used for data visualization?

Popular Tools:

- Tableau
- Power BI
- Excel
- Matplotlib
- Seaborn
- Plotly

Purpose:

Convert raw data into understandable visuals.

80. What are challenges in Big Data visualization?

Challenges:

1. Large Data Volume

Huge datasets are difficult to visualize.

2. High Velocity

Real-time data updates are challenging.

3. Complexity

Complex relationships are hard to represent.

4. Performance Issues

Visualization tools may become slow.

5. Data Quality Problems

Poor-quality data affects visualization accuracy.

Solution:

Use scalable visualization tools and proper preprocessing.

80. What are challenges in Big Data visualization?

Big Data visualization becomes difficult because of the massive size and complexity of data.

Challenges:**1. Large Volume of Data**

Huge datasets require high storage and processing power.

2. High Velocity

Real-time streaming data changes rapidly.

3. Variety of Data

Data comes in multiple formats:

- Text
- Images
- Videos
- Sensor data

4. Performance Issues

Visualization tools may slow down with large datasets.

5. Data Quality Issues

Missing and inconsistent data affect visualization accuracy.

6. Complexity

Complex relationships are difficult to represent visually.

Solutions:

- Data reduction
- Sampling
- Distributed computing
- Advanced visualization tools

81. What is Big Data?

Big Data refers to extremely large and complex datasets that cannot be processed using traditional systems.

Characteristics of Big Data (5 Vs):

1. Volume

Large amount of data.

2. Velocity

Fast generation of data.

3. Variety

Different forms of data.

4. Veracity

Data quality and reliability.

5. Value

Useful insights from data.

Examples:

- Social media data
 - Banking transactions
 - IoT sensor data
-

82. What is Hadoop?

[Apache Hadoop](#) is an open-source framework used for distributed storage and processing of Big Data.

Main Components:

1. HDFS

Hadoop Distributed File System.

2. MapReduce

Processes data in parallel.

3. YARN

Resource management system.

Advantages:

- Scalable
 - Fault tolerant
 - Cost-effective
-

83. What is Hadoop ecosystem?

The Hadoop ecosystem is a collection of tools used with Hadoop.

Important Tools:

HDFS

Storage system.

MapReduce

Processing framework.

Hive

SQL-like querying tool.

Pig

Scripting language for data analysis.

HBase

NoSQL database.

Sqoop

Transfers data between Hadoop and databases.

Spark

Fast data processing engine.

84. What is MapReduce?

MapReduce is a programming model used for processing large datasets in parallel.

Two Phases:

1. Map Phase

Splits data into key-value pairs.

2. Reduce Phase

Combines and summarizes results.

Example:

Counting word frequencies in documents.

Advantages:

- Parallel processing
 - Scalability
 - Fault tolerance
-

85. What is Pig?

[Apache Pig](#) is a high-level platform for analyzing large datasets in Hadoop.

Features:

- Uses Pig Latin language
- Simplifies MapReduce coding
- Handles structured and unstructured data

Applications:

- Log processing
 - ETL operations
-

86. What is Hive?

[Apache Hive](#) is a data warehouse tool used for querying data stored in Hadoop.

Features:

- SQL-like language called HiveQL
- Easy querying of Big Data
- Supports large-scale analysis

Applications:

- Data summarization
 - Business intelligence
-

87. What is a Line Plot?

A Line Plot is a graph that displays data points connected by straight lines.

Uses:

- Showing trends over time
- Comparing continuous data

Example:

Monthly sales growth.

Advantages:

- Easy trend analysis
 - Simple visualization
-

88. What is a Scatter Plot?

A Scatter Plot displays relationships between two variables.

Features:

- Each point represents one observation
- Helps identify correlation

Applications:

- Relationship analysis
- Outlier detection

Example:

Height vs weight analysis.

89. What is a Histogram?

A Histogram represents frequency distribution of continuous data.

Features:

- Data divided into bins
- Shows data distribution

Applications:

- Identifying skewness
- Understanding spread of data

Example:

Marks distribution of students.

90. What is a Box Plot?

A Box Plot shows:

- Median
- Quartiles
- Outliers
- Distribution spread

Components:

- Minimum
- Q1

- Median
- Q3
- Maximum

Uses:

- Outlier detection
 - Comparing datasets
-

91. Which Python libraries are required for data wrangling?

Important Libraries:

Pandas

Data manipulation.

NumPy

Numerical operations.

OpenPyXL

Excel file handling.

Dask

Large-scale data processing.

PySpark

Distributed data processing.

92. How do you import libraries in Python?

Libraries are imported using the import keyword.

Example:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Purpose:

Access functions and tools from libraries.

93. From where did you collect your dataset?

Possible sources include:

- Kaggle

- UCI Machine Learning Repository
- Government Open Data Portals
- APIs
- Surveys
- Sensors
- IoT devices

Example Answer:

“The dataset was collected from Kaggle for educational and analytical purposes.”

94. How do you load a dataset into a pandas DataFrame?

Using pandas functions like `read_csv()`.

Example:

```
import pandas as pd
```

```
df = pd.read_csv('data.csv')
```

Other Functions:

- `read_excel()`
 - `read_json()`
-

95. What is a DataFrame?

A DataFrame is a two-dimensional tabular data structure in pandas.

Features:

- Rows and columns
- Labeled axes
- Supports multiple data types

Example:

Name	Age
A	20
B	25

96. How do you check missing values using `isnull()`?

Using pandas:

Example:

```
df.isnull()
```

To count missing values:

```
df.isnull().sum()
```

Purpose:

Detect incomplete data.

97. What is the use of describe() function?

The describe() function provides statistical summary of numerical columns.

Example:

```
df.describe()
```

It shows:

- Mean
 - Median
 - Standard deviation
 - Minimum
 - Maximum
 - Quartiles
-

98. How do you find the dimensions of a dataset?

Using:

```
df.shape
```

Output:

(Number of rows, Number of columns)

Example:

```
(100, 5)
```

Means:

- 100 rows
 - 5 columns
-

99. What are different types of variables in a dataset?

1. Numerical Variables

Contain numbers.

Examples:

Age, salary

2. Categorical Variables

Contain categories.

Examples:

Gender, city

3. Continuous Variables

Can take any value.

Example:

Height

4. Discrete Variables

Fixed count values.

Example:

Number of students

100. How do you check data types of variables?

Using:

`df.dtypes`

Example Output:

- `int64`
- `float64`
- `object`

Purpose:

Understand dataset structure.

101. What is data formatting?

Data formatting converts data into a consistent structure.

Examples:

- Standard date format
- Currency formatting
- Text formatting

Importance:

Improves readability and analysis accuracy.

102. What is data normalization?

Normalization scales numerical values into a fixed range.

Common Range:

0 to 1

Formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Benefits:

- Faster model training
 - Better performance
-

103. Why is normalization required?

Normalization is required because machine learning algorithms perform better when features are on similar scales.

Advantages:

- Prevents bias
- Improves convergence speed
- Better distance calculations

Example:

Salary values may dominate age values without normalization.

104. How do you convert data types in Python?

Using `astype()`.

Example:

```
df['Age'] = df['Age'].astype(int)
```

Common Conversions:

- int
 - float
 - string
 - datetime
-

105. What are categorical variables?

Categorical variables contain labels or categories instead of numerical values.

Examples:

- Gender
- City
- Department

Types:**Nominal**

No order.

Ordinal

Ordered categories.

106. How do you convert categorical variables into numerical variables?

Using encoding techniques.

Methods:**1. Label Encoding**

Assigns numeric labels.

2. One-Hot Encoding

Creates binary columns.

Example:

```
pd.get_dummies(df['City'])
```

107. What is encoding?

Encoding converts categorical data into numerical format so machine learning algorithms can process it.

Types:

- Label Encoding
 - One-Hot Encoding
 - Ordinal Encoding
-

108. What is the difference between label encoding and one-hot encoding?

Label Encoding

Assigns unique numbers

Suitable for ordinal data

Simple implementation

One-Hot Encoding

Creates separate binary columns

Suitable for nominal data

Avoids false ordering

Example:

Label Encoding:

Red = 1
Blue = 2

One-Hot Encoding:

Red Blue

1 0

0 1

109. How do you detect missing values in a dataset?

Using pandas functions.

Example:

`df.isnull()`

To count:

`df.isnull().sum()`

Purpose:

Identify incomplete records.

110. What are different methods to handle missing data?

Methods:**1. Deletion**

Remove rows or columns.

2. Mean Imputation

Replace with average value.

3. Median Imputation

Useful for skewed data.

4. Mode Imputation

For categorical variables.

5. Forward Fill

Use previous value.

6. Backward Fill

Use next value.

111. What are inconsistencies in data?

Inconsistencies are errors or mismatched formats in datasets.

Examples:

- Different spellings
- Mixed date formats
- Duplicate entries

Problems caused:

- Incorrect analysis
 - Poor model performance
-

112. How do you detect inconsistencies?**Methods:**

- Checking duplicates
- Data validation
- Pattern matching
- Range checking

Example:

Different spellings:

- Mumbai
- mumbai
- MUMBAI

113. What are outliers?

Outliers are extreme values significantly different from other observations.

Example:

Most salaries:

20k–50k

One salary:

10 lakh

That value is an outlier.

Causes:

- Errors
- Rare events
- Natural variation

114. How do you detect outliers?

Methods:

1. Box Plot

Visual method.

2. Z-score Method

Formula

$$z = \frac{x - \mu}{\sigma}$$

x

μ

σ

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

3. IQR Method

Formula

$$IQR = Q3 - Q1$$

115. What techniques are used to handle outliers?

Techniques:

1. Remove Outliers

Delete extreme values.

2. Capping

Limit extreme values.

3. Transformation

Apply log transformation.

4. Winsorization

Replace outliers with nearest values.

116. What is data transformation?

Data transformation converts data into a suitable format for analysis.

Examples:

- Scaling
- Encoding
- Aggregation
- Log transformation

Benefits:

- Better model performance
 - Easier analysis
-

117. Why is transformation applied?

Transformation improves:

- Data quality
- Accuracy
- Model performance

Reasons:

- Reduce skewness
 - Normalize data
 - Convert non-linear relationships
-

118. What is skewness?

Skewness measures asymmetry of data distribution.

Types:

Positive Skew

Tail on right side.

Negative Skew

Tail on left side.

Zero Skew

Symmetrical distribution.

119. How do you reduce skewness in data?

Techniques:

1. Log Transformation

`np.log(data)`

2. Square Root Transformation

3. Box-Cox Transformation

Purpose:

Make data more normally distributed.

120. What is normalization vs standardization?

Normalization

Scales between 0 and 1

Uses min-max scaling

Sensitive to outliers

Standardization

Centers around mean

Uses z-score

Less sensitive

Standardization Formula

120. What is normalization vs standardization?

Normalization and Standardization are feature scaling techniques used in data preprocessing to improve machine learning model performance.

Normalization

Normalization scales data into a fixed range, usually between 0 and 1.

Formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Where:

- x = original value
 - x_{min} = minimum value
 - x_{max} = maximum value
-

Characteristics of Normalization

- Values remain within fixed range
 - Sensitive to outliers
 - Useful when data has no normal distribution
-

Example

Suppose:

- Minimum = 10
- Maximum = 50
- Value = 30

Normalized value:

$$(30 - 10) / (50 - 10) = 20/40 = 0.5$$

Applications

- Neural Networks
 - Image Processing
 - Distance-based algorithms
-

Standardization

Standardization transforms data so that:

- Mean = 0
- Standard Deviation = 1

It is also called Z-score normalization.

Formula

$$z = \frac{x - \mu}{\sigma}$$
$$z = \frac{x - \mu}{\sigma} \approx 1.2$$
$$\Phi(z) \approx 88.5\%$$

Where:

- x = original value
 - μ = mean
 - σ = standard deviation
-

Characteristics of Standardization

- Data is centered around mean
 - Less affected by outliers
 - Produces negative and positive values
-

Applications

- Logistic Regression
 - SVM
 - PCA
 - K-Means Clustering
-

Difference Between Normalization and Standardization

Normalization

Scales data between 0 and 1

Standardization

Centers data around mean

Normalization	Standardization
Uses min-max scaling	Uses z-score
Sensitive to outliers	Less sensitive
Suitable for bounded data	Suitable for normally distributed data

121. How do you convert non-linear data into linear form?

Non-linear data can be converted into linear form using transformation techniques.

This helps machine learning models understand relationships more effectively.

Methods to Convert Non-Linear Data into Linear Form

1. Logarithmic Transformation

Used when data grows exponentially.

Formula

$$y = \log(x)$$

Example:

Income distribution often becomes more linear after log transformation.

2. Square Root Transformation

Used for moderately skewed data.

Formula

$$y = \sqrt{x}$$

3. Polynomial Transformation

Adds polynomial terms to create linear relationships.

Example Formula

$$y = ax^2 + bx + c$$

a

b

c

4. Box-Cox Transformation

Used to stabilize variance and normalize data.

5. Feature Engineering

Creating new features from existing data.

Example:

Area = Length × Width

Importance of Converting Non-Linear Data

- Improves model accuracy
 - Better interpretation
 - Reduces skewness
 - Helps linear algorithms perform better
-

Example

Suppose house prices increase exponentially with area.

Applying logarithmic transformation may convert the relationship into a linear one suitable for Linear Regression.

122. Why is scaling important in data analysis?

Scaling is important because machine learning algorithms are sensitive to feature magnitudes.

If one feature has very large values, it may dominate smaller-valued features.

Why Scaling is Needed

1. Improves Machine Learning Performance

Algorithms perform better when features are on similar scales.

2. Faster Convergence

Gradient-based algorithms train faster after scaling.

3. Prevents Feature Dominance

Example:

- Salary = 500000
- Age = 22

Without scaling, salary dominates age.

4. Improves Distance Calculations

Algorithms like:

- KNN
- K-Means
- SVM

depend on distance measurements.

Scaling ensures fair distance calculation.

Types of Scaling

1. Normalization

Scales data between 0 and 1.

Formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

2. Standardization

Centers data around mean.

Formula

$$z = \frac{x - \mu}{\sigma}$$
$$z = \frac{x - \mu}{\sigma} \approx 1.2$$
$$\Phi(z) \approx 88.5\%$$

Algorithms That Require Scaling

- KNN
 - K-Means
 - SVM
 - Logistic Regression
 - Neural Networks
 - PCA
-

Algorithms Less Affected by Scaling

- Decision Tree
 - Random Forest
 - XGBoost
-

Example

Suppose:

- Height values: 150–190
- Salary values: 20,000–2,00,000

Without scaling, salary heavily influences calculations.

After scaling, both features contribute equally.